Titanic Exploratory Data Analysis (EDA)

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1. Introduction

Exploratory Data Analysis (EDA) is the process of examining and visualizing data to understand its structure, identify patterns, detect anomalies, and uncover relationships between variables. In this project, EDA is performed on the Titanic dataset to analyze passenger characteristics and determine the factors that influenced survival during the Titanic disaster.

2. Objective

The primary objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to gain a deeper understanding of passenger characteristics and survival patterns.

The specific objectives are:

- To understand the structure and quality of the dataset.
- To identify missing values and data inconsistencies.
- To analyze passenger demographics such as age and gender.
- To examine the relationship between passenger class, fare, and survival.
- To identify trends, patterns, correlations, and outliers through visualizations.

3. About the Dataset

The Titanic dataset contains passenger information collected from individuals who traveled aboard the RMS Titanic. The dataset includes various features such as passenger class, age, gender, family information, ticket fare, embarkation port, and survival status.

For this Exploratory Data Analysis project, only the train.csv file has been used.

Why Only train.csv?

The train.csv dataset contains the target variable Survived, which indicates whether a passenger survived (1) or did not survive (0) the disaster. Since the objective of this project is to analyze survival patterns and understand the factors affecting passenger survival, the availability of the Survived column is essential.

The Titanic dataset also includes:

- test.csv – Contains passenger information but does not include the Survived column. It is primarily used for making predictions after training machine learning models.
- gender_submission.csv – A sample submission file provided by Kaggle that demonstrates the required format for prediction submissions.

Since this project focuses exclusively on Exploratory Data Analysis rather than predictive modeling, the train.csv file is sufficient and most appropriate for the analysis.

4. Data Understanding

The Titanic dataset contains information about passengers aboard the RMS Titanic and includes demographic details, ticket information, passenger class, fare, and survival status.

For this analysis, the **train.csv** file containing **891 records** and **12 features** was used.

Key Features

Feature	Description
Survived	Survival Status (0 = No, 1 = Yes)
Pclass	Passenger Class
Sex	Passenger Gender
Age	Passenger Age
SibSp	Number of Siblings/Spouses
Parch	Number of Parents/Children
Fare	Ticket Fare
Embarked	Port of Embarkation

The dataset contains both numerical and categorical variables, making it suitable for statistical and visual analysis.

5. Data Quality Assessment

Data quality checks were performed to identify missing values and duplicate records.

Missing Values

Column	Missing Values
Age	177
Cabin	687
Embarked	2

Duplicate Records

- Duplicate Records Found: **0**

Observations

- The **Cabin** column contains the highest number of missing values.
- The **Age** column contains a moderate number of missing entries.
- No duplicate records were found in the dataset.

6. *Exploratory Data Analysis*

6.1 Survival Distribution Analysis

The target variable Survived was analyzed to understand the overall survival distribution among passengers.

Findings

- Total Passengers: 891
- Survived: 342
- Did Not Survive: 549

Observation

Approximately 61.6% of passengers did not survive, while 38.4% survived.

6.2 Gender Analysis

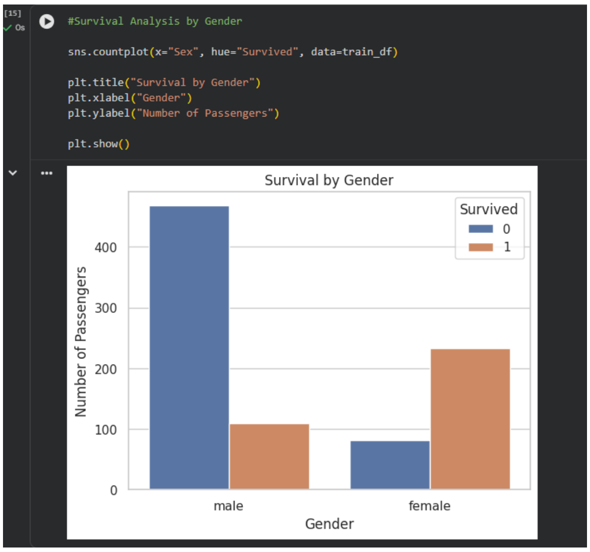
Passenger survival was analyzed based on gender.

Survival Rate by Gender

Gender	Survival Rate
Female	74.20%
Male	18.89%

Observation

Female passengers had a significantly higher survival rate than male passengers, indicating that gender strongly influenced survival outcomes.

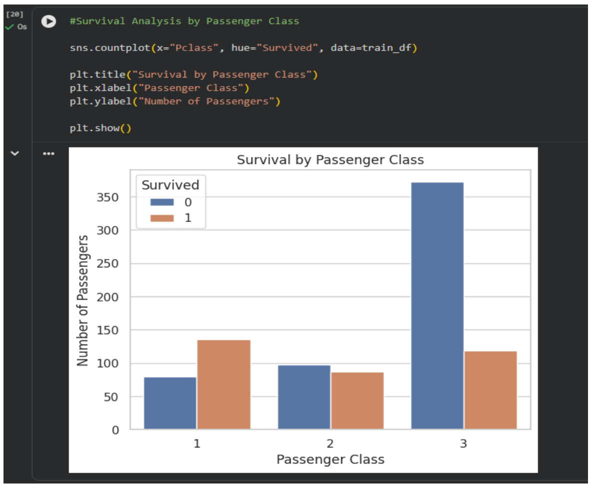


6.3 Passenger Class Analysis

Passenger class was analyzed to understand its impact on survival.

Survival Rate by Class

Class	Survival Rate
First Class	62.96%
Second Class	47.28%
Third Class	24.24%



Observation

First-Class passengers had the highest survival rate, while Third-Class passengers had the lowest survival rate, suggesting that travel class played an important role in survival.

6.4 Age Analysis

The Age variable was analyzed to understand passenger demographics and age distribution.

Findings

- Average Age: 29.7 years
- Median Age: 28 years
- Youngest Passenger: 0.42 years
- Oldest Passenger: 80 years

Observation

Most passengers were between **20 and 40 years of age**. A few elderly passengers appeared as outliers, but Age showed only a weak relationship with survival.

6.5 Fare Analysis

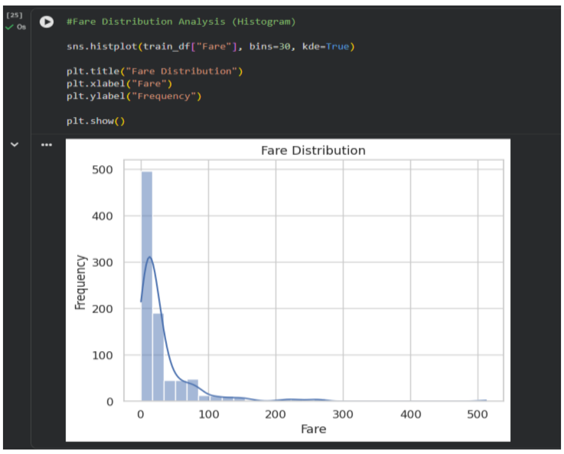
The Fare variable was analyzed to understand ticket pricing patterns and identify outliers.

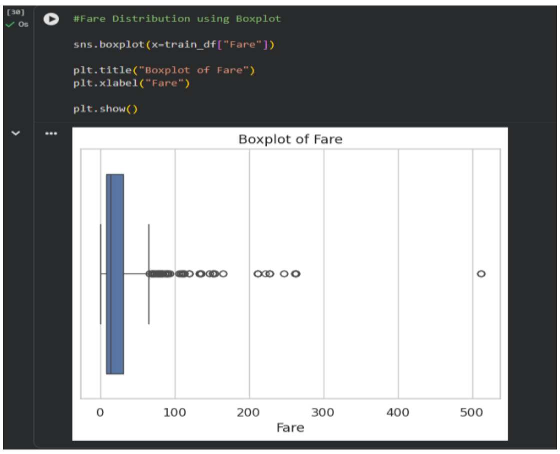
Findings

- Average Fare: 32.20
- Median Fare: 14.45
- Maximum Fare: 512.33

Observation

The Fare distribution was highly right-skewed. Most passengers paid relatively low fares, while a small number of passengers paid exceptionally high fares. Higher fares were generally associated with better survival outcomes.





6.6 Embarkation Port Analysis

The Embarked feature was analyzed to understand passenger boarding locations and their survival rates.

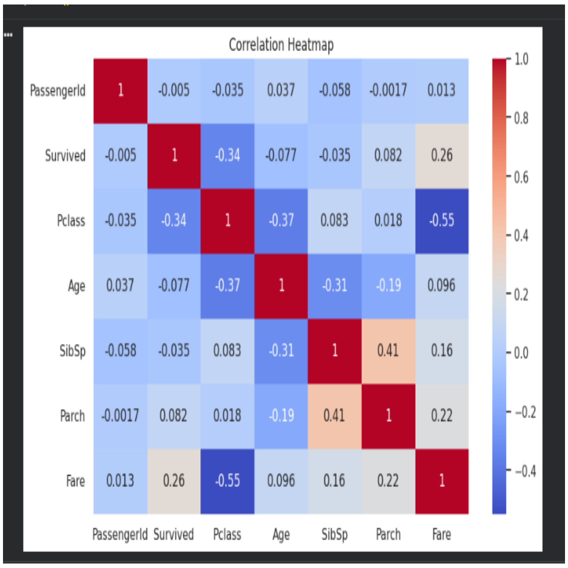
Passenger Distribution

Port	Passengers
Southampton (S)	644
Cherbourg (C)	168
Queenstown (Q)	77

Observation

Southampton was the primary embarkation port for most passengers. Cherbourg recorded the highest survival rate, which may be related to the larger proportion of First-Class passengers boarding from this port.

6.7 Correlation Heatmap Analysis



A correlation heatmap was generated to identify relationships between numerical variables in the Titanic dataset. Correlation values range from -1 to +1, where positive values indicate a direct relationship and negative values indicate an inverse relationship.

Key Findings

Variables	Correlation
Survived & Pclass	-0.34
Survived & Fare	0.26
Pclass & Fare	-0.55
SibSp & Parch	0.41

Observation

The heatmap revealed that **Passenger Class (Pclass)** and **Fare** had the strongest relationship with survival. First-Class passengers generally paid higher fares and had better survival rates. Age showed only a weak relationship with survival.

6.8 Pairplot Analysis

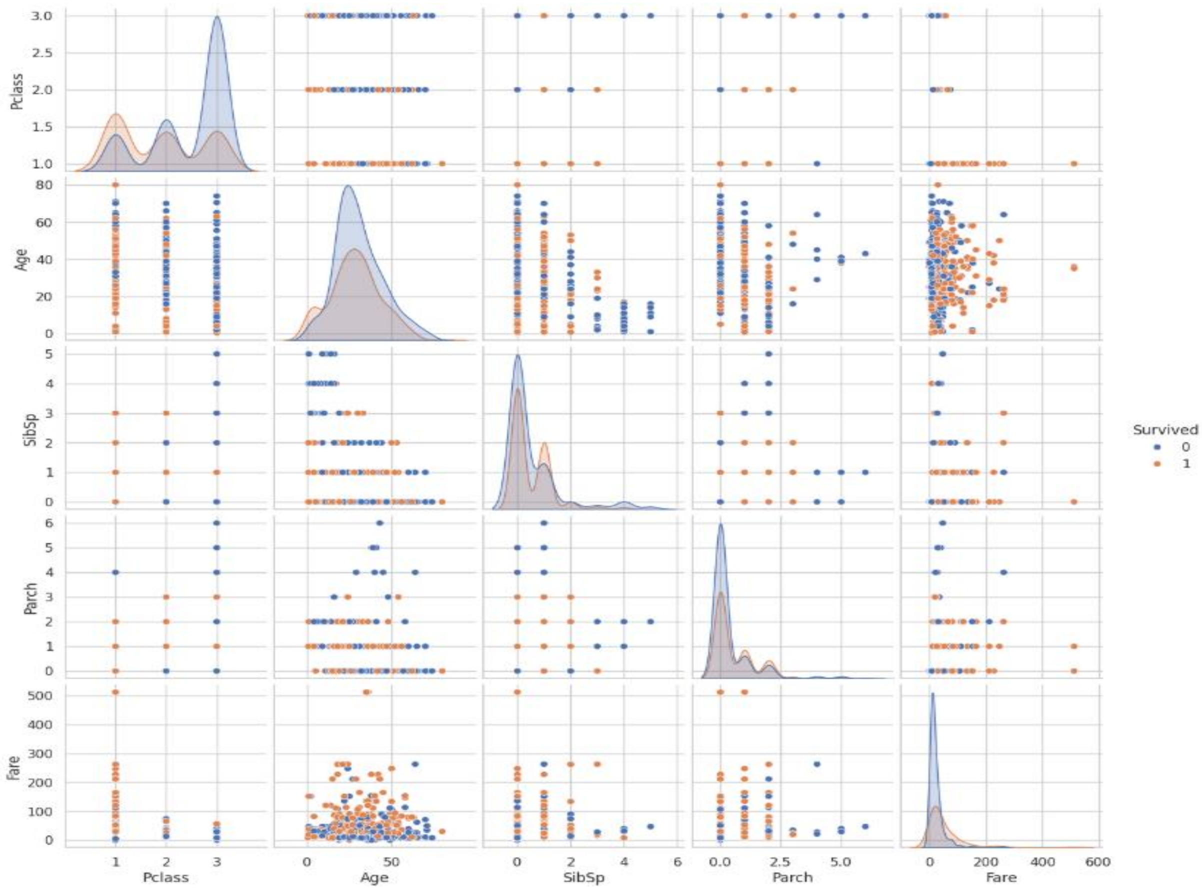
A pairplot was used to visualize relationships among multiple numerical variables, including Age, Fare, Passenger Class, and Survival.

Findings

- Higher fares were associated with higher survival rates.
- First-Class passengers generally paid higher fares.
- Age showed no strong separation between survivors and non-survivors.
- Family-related features displayed moderate clustering patterns.

Observation

The pairplot confirmed that Fare and Passenger Class were more influential in determining survival outcomes than Age.



6.9 Scatter Plot Analysis (Age vs Fare)

A scatter plot was created to examine the relationship between passenger Age and Fare while distinguishing survival outcomes.

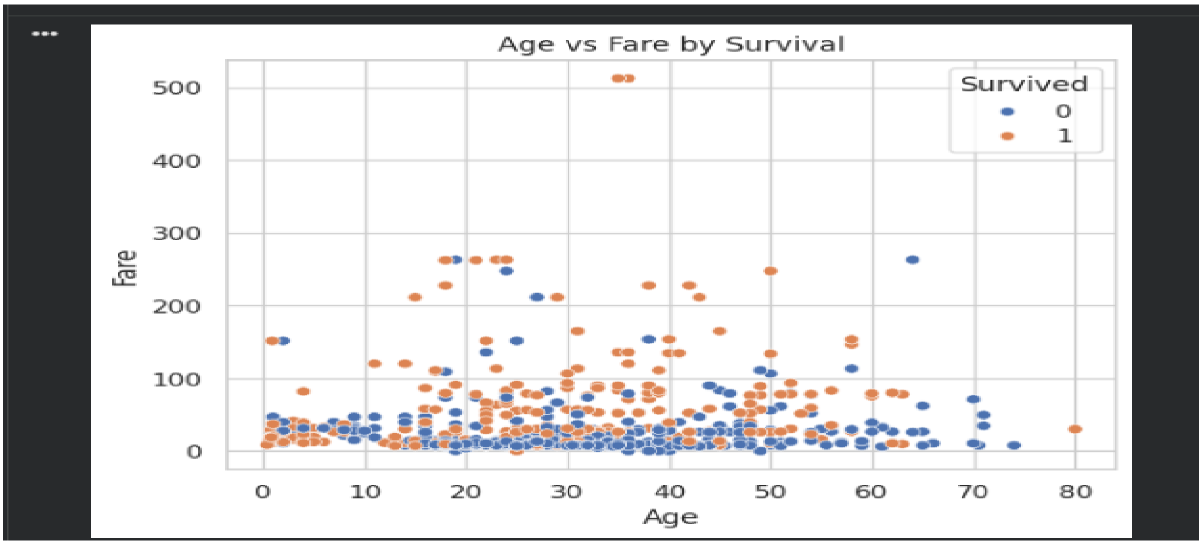
Findings

- Most passengers paid relatively low fares.
- Passengers paying higher fares were more likely to survive.

- No strong relationship was observed between Age and Fare.
- Survival patterns appeared more closely related to Fare than Age.

Observation

The scatter plot further supports the conclusion that ticket fare had a greater impact on survival than passenger age.



7. Key Insights

- Female passengers had significantly higher survival rates than male passengers.
- First-Class passengers experienced the highest survival rates.
- Passenger Class and Fare were the strongest factors associated with survival.
- Most passengers were between 20 and 40 years of age.
- The Fare distribution was highly right-skewed and contained several outliers.
- Southampton was the primary embarkation port for most passengers.
- Missing values were mainly concentrated in the Cabin and Age columns.

8. Recommendations

- Passenger Class and Fare should be considered important predictors in survival analysis.
- Missing values in Age and Cabin should be handled before building predictive models.
- Additional feature engineering could improve future machine learning performance.
- Gender and Passenger Class appear to be the most influential survival indicators.

9. Conclusion

This Exploratory Data Analysis of the Titanic dataset successfully identified important factors influencing passenger survival. Through statistical analysis and visualizations, meaningful patterns, trends, and relationships were uncovered.

The analysis demonstrated that **Gender, Passenger Class, and Fare** were the most significant variables affecting survival outcomes. Female passengers and First-Class travelers exhibited substantially higher survival rates, while Fare showed a positive relationship with survival probability.

Overall, the project highlights the importance of Exploratory Data Analysis in understanding datasets, identifying data quality issues, and extracting actionable insights before applying predictive modeling techniques.